

Non-identifiability of deep networks and the rate results of Kohler & Langer (2021): a commentary on the theorems and their proofs

Commentary on: M. Kohler and S. Langer, “On the rate of convergence of fully connected deep neural network regression estimates,” *Ann. Statist.* 49(4):2231–2249, 2021, together with Supplements A and B

Abstract

The map from weight vectors to the functions realized by a fully connected ReLU network is not injective—its fibers contain positive-dimensional manifolds—and its image is not closed. Deep networks are therefore *non-identifiable*: the weights cannot be recovered from the realized function, the least squares problem over weights has non-isolated (possibly non-existent) minimizers, and weight-norm-based complexity measures are ill-posed. This note examines, lemma by lemma, where this could obstruct the results of Kohler and Langer and how the paper avoids it. Three mechanisms carry the argument. *(i)* The statistical analysis (Lemma 18 and Lemma 19 of Supplement B) lives entirely on the realized, truncated function class: the oracle inequality uses only the basic inequality satisfied by *any* (near-)empirical-risk minimizer—no localization at a parameter, no curvature—and complexity is measured by VC bounds that are constant on the fibers of the parametrization and blind to weight magnitudes. *(ii)* The approximation theory (Theorems 2–3 and Supplement A) is purely existential; it needs one witness per fiber, never a canonical one. *(iii)* Far from being a nuisance, the representational redundancy is *spent as a resource*: zero-padding, exact identity subnetworks, composition by “weight melting,” storage of Taylor data directly in weights, and a base- B digit-packing scheme whose weights are exponentially large in M^d . We quantify why these devices are necessary: a naive fully connected construction only achieves the rate $n^{-p/(p+d)}$, and the constructions that restore $n^{-2p/(2p+d)}$ are admissible precisely because the analysis never attempts to identify parameters.

1 The realization map and its degeneracies

Fix a depth $L \in \mathbb{N}$ and a width $r \in \mathbb{N}$ and let $\mathcal{F}(L, r)$ be the class of fully connected ReLU networks defined in displays (1)–(2) of the paper. A network is specified by a weight vector

$$\theta = (c_{i,j}^{(0)}, c_{i,j}^{(1)}, \dots, c_{1,i}^{(L)}) \in \mathbb{R}^W, \quad W = (d+1)r + (L-1)(r+1)r + (r+1),$$

the count used in the proof of Corollary 1 and in Lemma 19 of Supplement B. Write

$$\mathcal{R} = \mathcal{R}_{L,r} : \mathbb{R}^W \longrightarrow \mathcal{F}(L, r), \quad \theta \longmapsto f_\theta,$$

for the *realization map*. **Non-identifiability** is the failure of injectivity of $\mathcal{R}$; it holds here in a strong, structured form.

(N1) Permutation symmetry. Relabeling the r neurons of hidden layer s (permuting the rows of layer $s - 1$ and, consistently, the columns of layer s) leaves f_θ unchanged. The finite group $\prod_{s=1}^L S_r$ acts on every fiber $\mathcal{R}^{-1}(f)$; generically each function has at least $(r!)^L$ distinct weight representations from this source alone.

(N2) Positive rescaling (ReLU homogeneity). Since $\sigma(\lambda u) = \lambda \sigma(u)$ for all $\lambda > 0$, the substitution, for any hidden unit (s, i) and any $\lambda > 0$,

$$(c_{i,\cdot}^{(s-1)}, c_{\cdot,i}^{(s)}) \mapsto (\lambda c_{i,\cdot}^{(s-1)}, \lambda^{-1} c_{\cdot,i}^{(s)}) \quad (1)$$

leaves f_θ invariant. This defines an action of $(0, \infty)^{Lr}$ (one factor per hidden neuron), free at every θ whose neurons each carry a nonzero incoming and a nonzero outgoing weight. Hence the fiber through such a θ contains a smooth manifold of dimension Lr : minimizers of any criterion depending on θ only through f_θ are *never* isolated, and the empirical L_2 -risk is exactly constant along these orbits, so every notion of curvature transverse to nothing—Hessians, Fisher information—is degenerate in at least Lr directions.

(N3) Degenerate strata. The fiber dimension is not even constant: if a unit is inactive on the relevant domain (pre-activation ≤ 0 there), all of its outgoing weights are free; if two units compute proportional functions, only a linear combination of their outgoing weights matters. So $\mathcal{R}$ is non-injective even modulo the group generated by (N1)–(N2).

(N4) Non-properness and non-closed image. $\mathcal{R}$ is not proper, and $\mathcal{R}(\mathbb{R}^W)$ is in general not closed. A one-dimensional witness inside $\mathcal{F}(1, 2)$:

$$f_\lambda(x) = \lambda \sigma(x + \tfrac{1}{\lambda}) - \lambda \sigma(x) \xrightarrow{\lambda \rightarrow \infty} \mathbf{1}_{[0, \infty)}(x) \quad \text{pointwise, hence in } L_2(P_X) \text{ if } P_X(\{0\}) = 0, \quad (2)$$

while the limit, being discontinuous, is realized by no ReLU network. Weights diverge along the minimizing sequence while the realized functions converge outside the class; best approximations, and a fortiori exact empirical minimizers, need not exist (this is the general phenomenon proved by Petersen, Raslan and Voigtlaender [5]).

(N5) No gauge is fixed. Display (2) of the paper imposes *no* constraint on θ : no sparsity (unlike [6], where at most s weights are nonzero) and no magnitude bound (unlike $\max_j |w_j| \leq 1$ in [6]). In view of (1) this is coherent—weight magnitudes are not even well defined modulo $\mathcal{R}$ —but it makes the parameter space non-compact and rules out every entropy argument of the form “Lipschitz in θ on a bounded set.”

Two further levels of non-identifiability. *(N6) Data level.* The empirical risk in display (3) of the paper is a function of the value vector $(f(X_1), \dots, f(X_n))$ alone; all networks agreeing on the design are empirically indistinguishable, so even the realized *function* is not identified by the data. *(N7) Model level.* The hierarchical representation (7)–(8) of a regression function $m \in \mathcal{H}(l, \mathcal{P})$ is not unique: components $g_j^{(i)}, h_j^{(i)}$ can be reparametrized (rescaled, permuted, composed with invertible smoothness-preserving maps absorbed by the next level) without changing m ; the constituents are not identified by m .

What is identified. The estimand is well posed. By the Pythagorean identity

$$\mathbf{E} |f(X) - Y|^2 = \int |f - m|^2 P_X(dx) + \mathbf{E} |m(X) - Y|^2,$$

$m(x) = \mathbf{E}\{Y \mid X = x\}$ is the P_X -a.s. unique minimizer of the population risk over measurable functions. Theorem 1 of the paper is a statement exclusively about the well-posed functional $\int |m_n - m|^2 P_X(dx)$ of the *realized* estimate m_n , never about weights. The entire question of this commentary is how the proofs manage to stay on that side of the map $\mathcal{R}$.

2 Where non-identifiability could bite

For orientation we list the concrete obstructions, matched below to the devices that dispose of them.

- (C1) *Parameter asymptotics are impossible.* By (N1)–(N3), θ -consistency, $\sqrt{n}$ -normality, and any local analysis of the risk around “the” minimizer (Taylor expansion, invertible Hessian/Fisher information) are unavailable: minimizing sets are unions of orbits, and in the likelihood picture the model is *singular* in the sense of Watanabe [8] (the Kullback–Leibler divergence has non-isolated zeros). A classical M-estimation proof of a rate would be dead on arrival.
- (C2) *Existence of the estimator.* By (N4) the minimum in display (3) may not be attained; “the least squares estimate” may not exist, and if it exists it is a set, raising a measurable-selection issue.
- (C3) *Non-uniqueness at the function level.* By (N6), even granting existence, the empirical minimizer is a large equivalence class; a risk bound must not depend on which element is picked.
- (C4) *Complexity accounting.* With unbounded weights (N5), norm-based entropy bounds are ill-posed under (1); and since $\mathcal{R}$ is many-to-one, counting parameters might misstate the size of the realized class.
- (C5) *Which representation is being learned?* By (N7), “estimating the composition structure” is an unidentified inverse problem; a proof must not require the estimator (or the approximating network) to recover $g_j^{(i)}, h_j^{(i)}$.

3 The statistical proofs (Supplement B): function-level, uniform, gauge-invariant

The proof of Theorem 1 opens with the inequality (from Lemma 18 of Supplement B)

$$\begin{aligned} \mathbf{E} \int |m_n - m|^2 P_X(dx) \leq & \frac{c_{10}(\log n)^2 \left(\sup_{x_1^n} \log \mathcal{N}_1\left(\frac{1}{n c_3 \log n}, T_{c_3 \log n} \mathcal{F}(L_n, r_n), x_1^n\right) + 1 \right)}{n} \\ & + 2 \inf_{f \in \mathcal{F}(L_n, r_n)} \int |f - m|^2 P_X(dx), \quad (3) \end{aligned}$$

where $m_n = T_{c_3 \log n} \tilde{m}_n$ is the truncated least squares estimate. Every quantity on the right is a functional of the *set of realized functions* $\mathcal{F}(L_n, r_n)$ and of m ; the parameter θ has already disappeared. Three observations make this precise.

3.1 Only the basic inequality is used — (C1), (C2), (C3) neutralized

The only property of $\tilde{m}_n$ that the proof of Lemma 18 exploits (it descends from Theorem 1 of Bagirov, Clausen and Kohler via the supplement of [2]) is the *basic inequality*

$$\frac{1}{n} \sum_{i=1}^n |\tilde{m}_n(X_i) - Y_i|^2 \leq \inf_{f \in \mathcal{F}(L_n, r_n)} \frac{1}{n} \sum_{i=1}^n |f(X_i) - Y_i|^2 (+ \delta_n), \quad (4)$$

together with the bound $|m_n| \leq c_3 \log n$ furnished by truncation. Consequently:

(a) Any measurable selection from the (orbit-shaped) argmin satisfies (4) with $\delta_n = 0$, and the resulting risk bound is *selection-free*: non-uniqueness (C1), (C3) is irrelevant because the concentration step is a *uniform* bound over the whole class $T_{c_3 \log n} \mathcal{F}(L_n, r_n)$ (this is exactly what the covering number in (3) buys), never a local statement at a distinguished minimizer. The proof contains no expansion of the risk in θ , no curvature condition, no identifiability hypothesis—the three ingredients that (C1) poisons in classical parametric M-estimation, where local injectivity of $\theta \mapsto f_\theta$ is a standing assumption.

(b) Existence (C2) is handled by fiat and then discharged: the authors assume after display (3) that the minimum exists “for simplicity” and note explicitly that all results persist for any estimate minimizing the empirical risk *up to a small additional term*, i.e. for (4) with $\delta_n > 0$ of order $o(n^{-1})$ – $O(n^{-1})$. In view of (N4) this remark is not cosmetic: minimizing sequences can have $\|\theta\| \rightarrow \infty$ with realized functions leaving the class in the limit, so near-minimizers are the mathematically robust notion, and they always exist.

(c) The truncation $T_{c_3 \log n}$ is where the unboundedness caused by (N5) is paid for, once: it converts a class with unbounded envelope (weights are free, so $\sup_f \|f\|_\infty = \infty$) into one with envelope $c_3 \log n$, matching the sub-Gaussian tail granted by $\mathbf{E} \exp(c_1 Y^2) < \infty$. Crucially, truncation acts on the *function*, not the weights: no gauge is fixed on $\mathbb{R}^W$.

3.2 Complexity is measured on the quotient — (C4) neutralized

The estimation term in (3) depends on the network class only through the $\|\cdot\|_1$ -covering number of the realized, truncated class—a quantity that is *constant on the fibers of $\mathcal{R}$* and invariant under the symmetry group of (N1)–(N2). Lemma 19 of Supplement B bounds it by the chain

$$\mathcal{N}_1(\epsilon, T_\beta \mathcal{F}(L, r), x_1^n) \leq 3 \left(\frac{4e\beta}{\epsilon} \log \frac{6e\beta}{\epsilon} \right)^V, \quad V = V_{T_\beta \mathcal{F}(L, r)^+} \leq V_{\mathcal{F}(L, r)^+} \leq c W L \log W, \quad (5)$$

the first inequality from Theorems 9.4/Lemma 9.2 of Györfi et al. [3], the last from Theorem 6 of Bartlett, Harvey, Liaw and Mehrabian [1], with $W = (d+1)r + (L-1)(r+1)r + r + 1$ as above; this yields $\log \mathcal{N}_1 \leq c_{56} \log(n) \log(Lr^2) L^2 r^2$ on the range $n^{-c_{55}} \leq \epsilon < c_{53} \log(n)/8$. Three structural points:

(a) *Gauge invariance, not gauge fixing.* The VC dimension of the subgraph class is a combinatorial functional of the set $\mathcal{F}(L, r)$; it does not see weight magnitudes at all. This is forced, not optional: the alternative route to entropy—discretizing the parameter space on a grid and pushing forward through $\mathcal{R}$, as is possible in [6] because there $|w_j| \leq 1$ —is unavailable here, since the estimator ranges over *all* of $\mathbb{R}^W$ and a cover of an unbounded set at fixed accuracy is infinite. One could not first prove that the minimizer lies in a bounded set either: by (1) every minimizer has orbit-mates of arbitrarily large norm. In short, (N2)+(N5) *force* a scale-free capacity measure, and (5) is exactly that. (Conversely, [6]’s bounded-weight gauge is a deliberate breaking of

the symmetry (1); both routes are legitimate, but they are not interchangeable—see §4.4 for why Kohler–Langer’s constructions could not survive the $|w_j| \leq 1$ gauge.)

(b) *Redundancy only makes the bound conservative.* Because $\mathcal{R}$ is many-to-one, a parameter-count bound like $cWL \log W$ can only *over-estimate* the capacity of the realized class: any shattering by realized functions lifts to parameters, never conversely. So non-identifiability cannot invalidate (5); the only risk is slack. And the slack from the known symmetries is negligible: the orbits of (N1)–(N2) have dimension Lr (+ a finite group), which is lower order than $W \asymp Lr^2$; quotienting them out would not change the order $WL \log W$. (That this order is essentially unimprovable— $V = \Omega(WL \log(W/L))$ —is shown in [1] by bit-extraction constructions of exactly the flavor the paper itself uses in Theorem 2(b); see §4.4. The two sides of the ledger are consistent.)

(c) *Uniformity over designs and over the class.* The $\sup_{x_1^n}$ in (3) and the class-level exponent V in (5) together give concentration *simultaneously* for every f in the class; this is what makes the bound indifferent to (N6): with the tuning of Theorem 1, $V_n \lesssim (\log n)^3 n^{1/(2\bar{p}/\bar{K}+1)} = o(n)$, the class is too small for design-interpolating pathologies, and *all* near-minimizers enjoy the same risk bound, whichever the algorithm returns.

3.3 Assembling Theorem 1: the two terms never mention θ

With $L_n = \lceil c_4 \log n \rceil$ and $r_n = \lceil c_5 \max_{(p,K) \in \mathcal{P}} n^{K/(2(2p+K))} \rceil$ (case (a)), (5) gives the estimation term (display (4) of the paper)

$$\frac{c_{10}(\log n)^2 \cdot c_{14}(\log n)^3 \log(c_{14}(\log n) n^{\frac{2}{2\bar{p}/\bar{K}+1}}) c_{14} n^{\frac{1}{2\bar{p}/\bar{K}+1}}}{n} \leq c_{15} (\log n)^6 n^{-\frac{2\bar{p}}{2\bar{p}+\bar{K}}},$$

where $(\bar{p}, \bar{K}) = \arg \min_{(p,K) \in \mathcal{P}} p/K$,

while the approximation term is bounded by Theorem 3(a) with $M_{j,i} = \lceil n^{1/(2(2p_j^{(i)}+K_j^{(i)}))} \rceil$ and $a_n = (\log n)^{1/(4(p_{\max}+1))}$, giving $c_{16}(\log n)^2 \max_{j,i} n^{-2p_j^{(i)}/(2p_j^{(i)}+K_j^{(i)})}$ after squaring the sup-norm bound (10). Both knobs—architecture (L_n, r_n) and grid sizes $M_{j,i}$ —are chosen at the level of function classes and approximation errors; no step asks which weights realize what. Case (b) is identical with the deep tuning $L_n = \lceil c_7 \max n^{K/(2(2p+K))} \log n \rceil$, $r_n = c_8$, and $L_n^2 r_n^2$ in (5) taking over the role of r_n^2 .

3.4 Representation-blindness of the approximation term — (C5) neutralized

The second term of (3) is an infimum over f ; it requires exhibiting *one* good network. Theorem 3 builds it along one hierarchical representation (6)–(8) of m , fixed arbitrarily. But the *value* of the infimum—hence the rate—does not depend on that choice; if m admits several representations with different smoothness/order profiles, Theorem 3 applies to each and the best resulting bound stands. Nothing in the estimator attempts to recover the components $g_j^{(i)}$ or $h_j^{(i)}$: the unidentified deconvolution of the composition (C5) is simply never posed. Two corollaries of this stance are worth recording: Remark 2 (rate optimality) compares against a *minimax lower bound over distributions* (Theorem 3 of [6]), a formulation intrinsically indifferent to parametrization; and Remark 3 (adaptation) selects width and depth by empirical risk on a hold-out sample—model selection, too, is performed purely on function values.

4 The approximation proofs (Supplement A): redundancy as a resource

Theorems 2 and 3 are existence statements: “there exists a network $\hat{f} \in \mathcal{F}(L, r)$ with $\|f - \hat{f}\|_{\infty, [-a, a]^d} \leq \dots$ ”. Non-uniqueness of representations is therefore harmless *a priori*—one witness per fiber suffices. What is striking in Supplement A is that the constructions go further and *consume* the redundancy of $\mathcal{R}$ as a budget. Five distinct mechanisms can be isolated.

4.1 Padding and identity lanes: choosing representatives of prescribed shape

Since $\mathcal{F}(L, r') \subseteq \mathcal{F}(L, r)$ for $r' \leq r$ (set superfluous weights to zero—the paper notes after (2) that the class “also contains networks with some weights chosen as zero”), subnetworks of unequal widths can be run *in parallel* in the same layers with widths adding; this is how $t_1 \in \mathcal{F}(l L_0, \max_i \sum_j r_j^{(i)})$ is assembled in the proof of Theorem 3(a). Dually, the *exact* identity network

$$\hat{f}_{\text{id}}(z) = \sigma(z) - \sigma(-z) = z, \quad \hat{f}_{\text{id}}^t = \text{id},$$

provides lossless depth padding: any already-computed scalar can be transported through further hidden layers at a cost of 2 neurons per layer and *zero* error. It is used ubiquitously “to synchronize the number of hidden layers” (proofs of Lemmas 7, 12, 16, 17; after (14)); in Theorem 3(b) it appears as permanent wiring— $2d$ neurons carry x and $2 \sum_{t \leq l-1} \tilde{N}_t$ neurons carry all previously computed $\hat{h}_j$ through every subsequent block (Fig. 5), which is precisely the term $2 \sum_{t=1}^{l-1} \tilde{N}_t + 2d$ in the width bound of Theorem 3(b)(ii). Both devices are, verbatim, choices of a representative of prescribed depth/width inside the fiber $\mathcal{R}^{-1}(f)$; they exist only because representations are massively non-unique. This is also what makes the homogeneous architecture $\mathcal{F}(L_n, r_n)$ of Theorem 1 usable at all: the heterogeneous approximants of Theorem 3 are embedded into it by padding.

4.2 Composition by weight melting

For $f \in \mathcal{F}(L_f, r_f)$ and $g \in \mathcal{F}(L_g, r_g)$ the paper composes networks so that $f \circ g \in \mathcal{F}(L_f + L_g, \max\{r_f, r_g\})$, one layer cheaper than the convention $f \circ \sigma(g)$ of [6]. The mechanism (Fig. 3): write g ’s affine output map as $z \mapsto Cz + c$ (in the last hidden activations z of g) and f ’s first pre-activation as $u \mapsto Au + b$; the junction of the composite is the single affine map

$$z \longmapsto (AC)z + (Ac + b),$$

i.e. the weights are “melted” into the product AC . Only that product is realized; the factorization is invisible ((A, C) and $(AT, T^{-1}C)$ melt identically). The construction thus *lives* on an equivalence class of parametrizations and simply picks the parsimonious representative—an operation that a framework insisting on identified parameters could not even express.

4.3 Theorem 2(a): weights as a lookup table, and the two-scale trick

The wide construction stores the target function *in the weights*. The sketch in the paper says it plainly: a layer pair of width $c_{22}M^d$ carries $c_{23}M^{2d}$ connections, and “each of the $c_{23}M^{2d}$ weights in our network is matched to one of the $c_{23}M^{2d}$ possible values of the derivatives of f .” Concretely

(proof of Lemma 3), with P_1, P_2 the coarse/fine partitions of (15),

$$\hat{\phi}_{3,1}^{(l,j)} = \sum_{i \in \{1, \dots, M^d\}} (\partial^l f)((\tilde{C}_{j,i})_{\text{left}}) \cdot \hat{f}_{\text{ind}, C_{i,1}}(x),$$

so the numbers $(\partial^l f)((\tilde{C}_{j,i})_{\text{left}})$, $\binom{d+q}{d} M^{2d}$ of them, *are* outgoing weights of indicator units. The two-scale detection—exact coarse indicators $\hat{f}_{\text{ind}} \in \mathcal{F}(2, 2d)$ (Lemma 6(a)) followed by the gated-value networks $\hat{f}_{\text{test}} \in \mathcal{F}(2, 2(2d+2))$ (Lemma 6(b)) on the data-dependent subcubes $A^{(j)}$ of (17)—selects the correct Taylor data on the fine grid using only $O(M^d)$ neurons. Everything is *exact* off the collars: for $x \in \bigcup_k (C_{k,2})_{1/M^{2p+2}}^0$ the recursion of Lemma 2 is reproduced without error, and the only inexact modules are the arithmetic ones— $\hat{f}_{\text{mult}} \in \mathcal{F}(R, 18)$ with $|\hat{f}_{\text{mult}}(x, y) - xy| \leq 2a^2 4^{-R}$ (Lemma 4) and the coefficient-fed polynomial net $\hat{f}_p$ (Lemma 5)—whose 4^{-R} errors are killed by $R = \lceil \log_4 M^{2p} \rceil$. Near the collars the value is not accurate but is *bounded* (end of Lemma 3), and the gate $\hat{f}_{\text{check}, P_2}$ (Lemma 10), the B-spline weight w_{P_2} of (22) realized in Lemma 9, and the 2^d shifted partitions with $\sum_v w_v \equiv 1$ on $[-a/2, a/2]^d$ (display (33)) convert the almost-everywhere bound into a supremum-norm bound. Three remarks tie this to identifiability:

(a) Every weight assignment above is by *fiat*; no step asks whether these weights could be recovered from the realized $\hat{f}$, and many manifestly could not (e.g. the weights feeding gated-off branches).

(b) Lemma 5’s polynomial network takes the Taylor *coefficients as inputs* $y_1, \dots, y_{\binom{d+q}{d}}$ rather than as weights: at that stage of the circuit, “parameters” of the local model travel on wires. The parameter/function boundary is deliberately porous—a design freedom that only exists because nothing downstream needs the boundary.

(c) The interval bookkeeping that pervades Supplement A (“the input of $\hat{f}_p$ is contained in the interval where (19) holds,” likewise (28), (38)) is the constructive counterpart of the statistical truncation of §3: since the arithmetic subnetworks are accurate only on compacta, all internal signals must be range-controlled; boundedness is engineered at the function level, never by constraining weights.

4.4 Theorem 2(b): digit packing, exponentially large weights, and why the gauge matters

The deep construction is the sharpest illustration. Depth replaces width: the fine-grid Taylor data is *generated*, not stored, by propagating along the enumeration $\tilde{C}_{1,i}, \dots, \tilde{C}_{M^d,i}$ of (34):

$$(\partial^l \hat{f})((\tilde{C}_{k+1,i})_{\text{left}}) = \sum_{\substack{j \in \mathbb{N}_0^d \\ \|j\|_1 \leq q - \|l\|_1}} \frac{(\partial^{l+j} \hat{f})((\tilde{C}_{k,i})_{\text{left}})}{j!} \tilde{v}_{k+1}^j + b_{k,i}^{(l)} \cdot c_{46} \left(\frac{2a}{M^2} \right)^{p - \|l\|_1},$$

with integer corrections $b_{k,i}^{(l)} \in \mathbb{Z}$, $|b_{k,i}^{(l)}| \leq e^d + 1$, chosen so that the propagated error stays below $c_{46}(2a/M^2)^{p - \|l\|_1}$, whence the final bias (36) of order $e^d c_{46}(2a/M^2)^p \asymp M^{-2p}$. The entire correction stream of a coarse cube is compressed into a *single real weight*

$$b_i^{(l)} = \sum_{k=1}^{M^d-1} (b_{k,i}^{(l)} + \lceil e^d \rceil + 2) B^{-k} \in [0, 1], \quad B = 4 + 2\lceil e^d \rceil,$$

and unpacked inside the network, one base- B digit per block of layers, by the floor network $\hat{f}_{\text{trunc}} \in \mathcal{F}(1, 2B)$ of Lemma 13, which is *exact* provided its input keeps distance $\geq 1/R$ from $\mathbb{N}$; Lemma 14 supplies exactly the required separation $B^{-(M^d-j-1)}$, so the slopes must be taken as

$$R_{M,i} = B^{M^d-i-1},$$

exponentially large in M^d .¹ Two consequences deserve emphasis.

(a) *Single parameters of unbounded information content.* One real weight carries $\Theta(M^d)$ digits; other weights (the $\hat{f}_{\text{trunc}}$ slopes) grow like B^{M^d} ; already in part (a) the indicator slopes are $B_M \geq M^{2p+2}$. Such representatives of the fiber exist *only* because (N5) fixes no gauge. Under the normalization $\max_j |w_j| \leq 1$ of [6] these networks are inadmissible, and reproducing them with bounded weights would require synthesizing the large constants from depth, changing the architecture counts. The freedom to be numerically extravagant is a direct dividend of refusing to identify (hence to normalize) parameters—and it is statistically free, because the entropy bound (5) is magnitude-blind (§3.2(a)).

(b) *Consistency with capacity.* Digit packing is the same bit-extraction mechanism that proves the near-matching *lower* bound $V = \Omega(WL \log(W/L))$ in [1]. The construction therefore operates at the information-theoretic capacity of the parameter budget; the paper’s rate computation shows the two sides balance exactly at $n^{-2p/(2p+K)}$.

4.5 Theorem 3: composition without deconvolution

Both proofs of Theorem 3 mirror one chosen representation of $m \in \mathcal{H}(l, \mathcal{P})$: subnetworks $\hat{f}_{\text{wide}, g_j^{(i)}}$ in parallel levels (part (a), Fig. 4) or $\hat{f}_{\text{deep}, g_j^{(i)}}$ in series with identity lanes (part (b), Fig. 5). The error control is the induction (44)/(46) of Supplement A, e.g.

$$|\hat{h}_j^{(i)}(x) - h_j^{(i)}(x)| \leq c_{50} \cdot i \cdot (K_{\max} C_{\text{Lip}})^{i-1} a^{4(p_{\max}+1)} \max_{j,i} M_{j,i}^{-2p_j^{(i)}},$$

which uses only two *functional* facts: the sup-norm accuracy of each block on a fixed compactum, and the Lipschitz continuity of the true $g_j^{(i)}$ to propagate perturbations of the *inputs*. No parameter of any block is tracked, matched, or recovered; stability of composition replaces identification of components. This is the proof-level answer to (C5).

4.6 The quantitative payoff, and why the tricks are necessary

Corollary 1 records the headline: sup-error ϵ on $[-a, a]^d$ with a fully connected network of $c_{19} \epsilon^{-d/(2p)}$ *total* parameters, i.e. error $W^{-2p/d}$ —a quadratic improvement over the $\epsilon^{-d/p}$ *nonzero* parameters of [2, 6] (Remark 5). This is not a curiosity but the precise price of abandoning identifiable structure (sparsity patterns) in favor of an over-complete parametrization, as the following back-of-envelope computation with (5) shows. Suppressing log factors and treating a single (p, d) component:

Naive fully connected one-scale construction. One localized unit group per cell of a single grid with mesh h : sup-error $\eta \asymp h^p = M_*^{-p}$ needs width $r \asymp M_*^d$, depth $O(\log)$, hence $W \asymp M_*^{2d}$ and

¹The display in the proof of Lemma 12 reads $R_{M,i} = 1/(4+2\lceil e^d \rceil)^{M^d-i-1}$; comparison with Lemma 13 (exactness requires $\min_{j \in \mathbb{N}} |z - j| \geq 1/R$) and Lemma 14 shows this is an evident misprint for the reciprocal, as stated here.

$V \asymp M_*^{2d}$. Balancing $V/n \asymp \eta^2$ gives

$$\eta^2 \asymp n^{-\frac{2p}{2p+2d}} = n^{-\frac{p}{p+d}},$$

strictly slower than the target for every $d \geq 1$.

The paper’s constructions. Part (a): width $r \asymp M^d$ but, via the two-scale trick, error M^{-2p} on the *fine* grid of M^{2d} cells; $W \asymp M^{2d}$, $V \asymp M^{2d}$, and balancing $M^{2d}/n \asymp M^{-4p}$ yields $\eta^2 \asymp n^{-2p/(2p+d)}$. Part (b): $r \asymp \text{const}$, $L \asymp M^d$, so $W \asymp M^d$ and $V \lesssim WL \log W \asymp M^{2d}$ —the same balance, the same rate, now with error $\asymp W^{-2p/d}$ per parameter.

In words: because the entropy (5) charges *all* W parameters of the fully connected class—redundant or not, in contrast to the $s \log(\cdot)$ accounting available to sparse classes—the constructions must double the approximation exponent per parameter, and the two-scale/digit-packing devices of §4.3–§4.4 do exactly that. Non-identifiability enters twice, with opposite signs, and cancels: the over-complete parametrization inflates the charged budget, and the unconstrained, non-canonical weight assignments (lookup tables, packed digits, exponential slopes) deflate the required budget by the same factor.

Finally, the comparison with [6] isolates a *different* species of non-identifiability that the paper eliminates by design. The sparse class of [6] is a union $\bigcup_{S:|S|\leq s} \mathcal{F}_S$ over supports; the estimator implicitly selects a support, and—as Section 1.4 of the paper stresses—“it was described how many weights are nonzero but not which of the weights are nonzero”: the *topology* of the optimal network is combinatorially ambiguous, an obstacle both to specification and to implementation. Kohler–Langer’s class has one fixed topology (hence the five-line Keras program of Listing 1); the only remaining ambiguity is the benign fiber structure of §1, which the function-level analysis of §3 never sees.

5 What the circumvention does not deliver

Because identifiability is avoided rather than restored, several questions remain outside the theorems—by design, but worth stating precisely. (i) *Computation.* Nothing is claimed about *finding* $\tilde{m}_n$: the empirical risk on $\mathbb{R}^W$ inherits the full symmetry group, so every minimizer comes with its $\prod_s S_r \times (0, \infty)^{L_r}$ -orbit and $(r!)^L$ permuted copies; the landscape analysis of gradient training is untouched by Theorem 1. (ii) *Inference on weights.* Consistency, standard errors, or interpretation of $\hat{\theta}$ are meaningless objects here; the singular-model machinery [8] that would be needed for likelihood asymptotics is entirely bypassed because no likelihood expansion is ever made. (iii) *Uniqueness.* m_n itself is not unique; only its risk bound is, uniformly over selections. (iv) *Adaptivity.* The tuning $(L_n, r_n, M_{j,i})$ uses $(p, K) \in \mathcal{P}$; Remark 3 restores practicality through split-sample selection—which, fittingly, is again a purely functional criterion (empirical risk on a hold-out), so that even model selection is performed identifiability-free.

6 Summary

Manifestation of non-identifiability	How the paper avoids or exploits it	Location
Orbits (N1)–(N3): no isolated minimizers, degenerate curvature	Risk bound for the realized function only; proof uses the basic inequality (4) + uniform concentration; no localization at a minimizer	Thm. 1; Lem. 18 (Supp. B)
Non-properness (N4): ERM may not exist	Near-minimizer slack δ_n suffices	Remark after (3); Lem. 18
No gauge (N5): unbounded weights, norm entropy ill-posed	Truncation $T_{c_3 \log n}$ at the function level; scale-free VC bound $V \leq cWL \log W$, invariant on fibers	Lem. 19 (Supp. B); [1]
Many-to-one parameter counting	Upper-bound direction only; orbit dimension $O(Lr) \ll W$	§3.2(b)
Design-level indistinguishability (N6)	Selection-free, class-uniform bounds with $V_n = o(n)$	(3), (5)
Representation ambiguity of $\mathcal{H}(l, \mathcal{P})$ (N7)	Existence-only approximation along an arbitrary representation; Lipschitz stability replaces deconvolution	Thm. 3; (44), (46)
Redundant representations of a given f	<i>Exploited:</i> zero-padding, identity lanes, weight melting, weights-as-lookup-table, digit packing with $R_{M,i} = B^{M^d-i-1}$	Supp. A; Figs. 3–5; Lems. 3–6, 12–14
Support ambiguity of sparse classes	Eliminated by fixing a fully connected topology; total-parameter budget made affordable by the $W^{-2p/d}$ approximation rate	§1.4; Thm. 2; Cor. 1

Verdict. Non-identifiability is a genuine obstruction only to questions the paper does not ask. For the question it does ask—the L_2 risk of the realized estimate—the proofs are organized so that every quantity that appears (empirical risk values, covering numbers, VC dimensions, sup-norm approximation errors, Lipschitz propagation constants) is a function of $\mathcal{R}(\theta)$, never of θ ; existence and uniqueness of the minimizer are traded for near-minimization and uniformity; and the freedom of the fibers is not merely tolerated but spent, as the construction budget that lets fully connected networks match—indeed, per parameter, quadratically beat—their sparse, partially identified competitors.

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